# Refraction

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect a change in wave speed to wavelength and direction.

**SUCCESS CRITERIA - I can...**

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “refraction” and “normal line” to explain your answer. Sentence starters are optional.

**refraction**: change in wave direction when speed changes at a boundary

**normal line**: a reference line perpendicular to a surface

**medium**: the material through which a mechanical wave travels

## 01 Warm-up: bring back an earlier idea

A wave snapshot has crests at **0.1**, **0.4**, **0.7 m**. **Find wavelength**. Its period is **0.05 s**; **find frequency**.

**Word help:** wavelength = distance between corresponding points on repeated waves

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

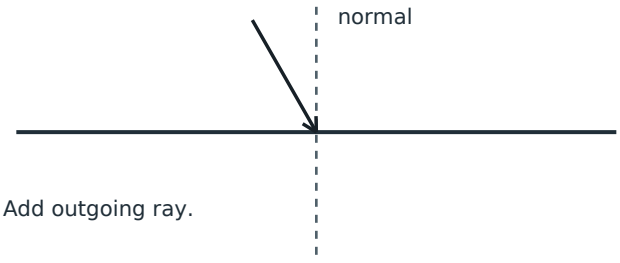
## 02 Worked example: explain the prediction

**Reference:** $n_1 \sin\theta_1 = n_2 \sin\theta_2$; $n = c/v$; $v = f\lambda$.
Angles from the normal. Stationary boundary: frequency stays constant.

**Worked reasoning:** Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

**Your turn:** Draw and **label the boundary**, normal, incident ray and refracted ray.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

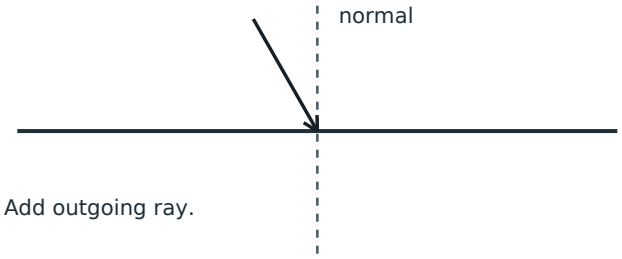



**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

**Mark the normal** at the boundary. A ray enters a slower medium at an angle. **Sketch the refracted direction**.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

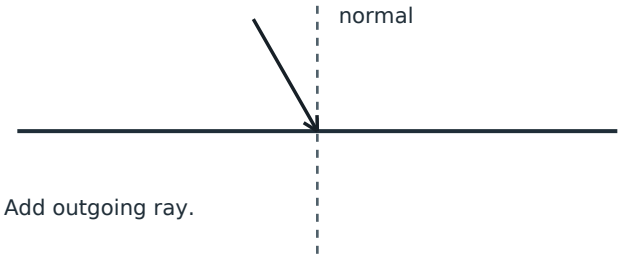



**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A laser ray travels from air (n = **1.00**) into glass (n = **1.50**) at **30°** to the normal. **Label both media** and angles, then use $n_1 \sin\theta_1 = n_2 \sin\theta_2$ to **find the refracted angle**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** ① ② ③

# Refraction

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is **20°**, **what angle** belongs in Snell's law?

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A **0.4 kg** metal cools from **92** to **32°C** while **0.3 kg** water warms from **12** to **32°C**. Use water c=**4200**; neglect cup/losses. **Find metal c**.
**Label temperatures**, samples and relevant energy transfers.

**Sentence starter:** I found … [unit]. This value means … in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

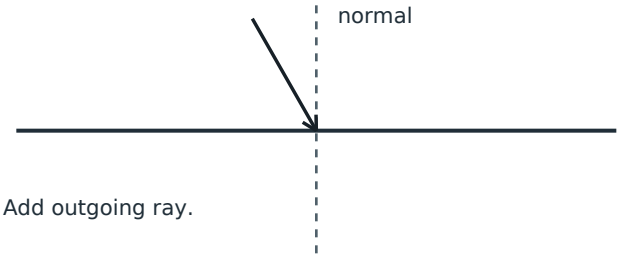
**Marzano check - circle:** ① ② ③

## 07 Exit check

From glass (n=**1.5**) to air, find critical angle using sin $\theta_c$=**1/1.5**. **State what happens above** that angle in this ideal case.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** I found … [unit]. This value means … in this situation.



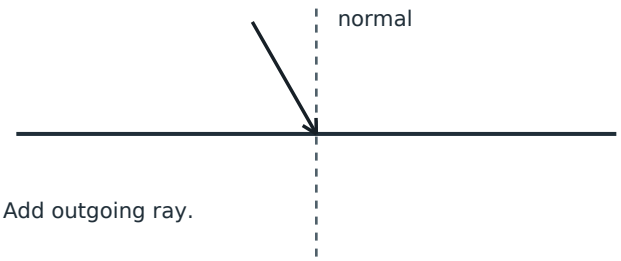

Add outgoing ray.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

A ray travels from water (n = **1.33**) into glass (n = **1.50**) at **40°** to the normal. Use $n_1 \sin\theta_1 = n_2 \sin\theta_2$ to **find the refracted angle**. Then **find the critical angle** for travel from this glass into water. **Label both cases**.

**Sentence starter:** I found … [unit]. This value means … in this situation.




Add outgoing ray.

**Marzano check - circle:** ① ② ③